

JEE-Main-09-04-2024 (Memory Based)
[EVENINGSHIFT]

Physics

Question: Dimensional formula of Planck's constant is

Options:

- (a) $[M^2L^2T^{-1}]$
- (b) $[M^1L^2T^{-1}]$
- (c) $[M^2L^2T^{-2}]$
- (d) $[ML^2T^{-3}]$

Answer: (b)

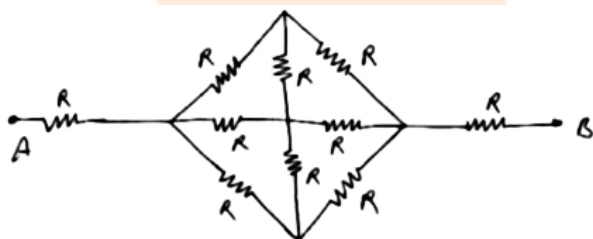
Question: Find the magnitude of force F, if the given system is in equilibrium

Options:

- (a) 10 N
- (b) $10/\sqrt{2}$
- (c) 0 N
- (d) $\frac{1}{10\sqrt{2}}N$

Answer: (a)

Question: The equivalent resistance between terminal A and B in the network shown



Options:

- (a) $4R/3$
- (b) $8R/3$
- (c) $3R$
- (d) $5R/2$

Answer: (b)

Question: The nuclei at rest breaks into two parts with mass ratio 1 : 2. The ratio of their velocity and direction is

Options:

- (a) Opposite Direction 2 : 1
- (b) Same Direction 1 : 2
- (c) Opposite Direction 1 : 1

(d) Same Direction 1 : 1

Answer: (a)

Question: Two cars A and B are moving towards each other with speed 20 m/s each. When 300 m apart, they both apply breaks which causes deceleration of 2 m/s^2 . The distance between them when they stop will be:

Options:

(a) 100 m

(b) 50 m

(c) 150 m

(d) 200 m

Answer: (a)

Question: For a wire, the original resistance was 50Ω at the initial temperature of 27°C . When the temperature is increased, its resistance becomes 62Ω . If the thermal coefficient of resistivity of the wire is $2.4 \times 10^{-2} \text{ K}^{-1}$, find the final temperature.

Options:

(a) 45°C

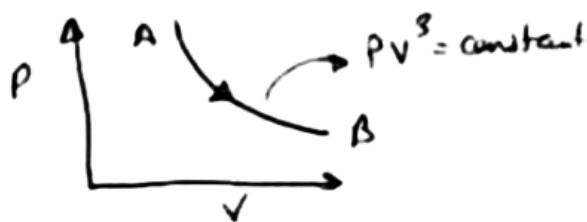
(b) 32°C

(c) 37°C

(d) 48°C

Answer: (c)

Question: Find the work done by a monatomic gas from A and B. Here the temperature of gas (1 mol) changes from 300 K to 330 K.



Options:

(a) 125 J

(b) 250 J

(c) 500 J

(d) 6250 J

Answer: (a)

Question: Two bubbles having radi r_A and r_B are having excess pressure P_A and P_B in them. If $P_A = 3P_B$, find r_A/r_B

Options:

(a) 9 : 1

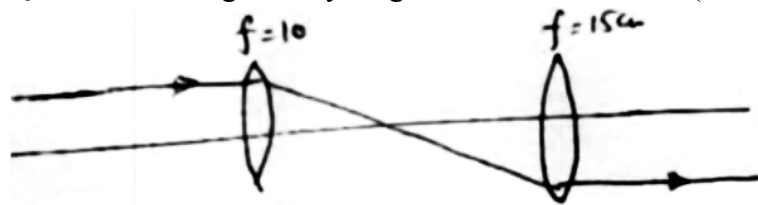
(b) 1 : 9

(c) 1 : 3

(d) 3 : 1

Answer: (c)

Question: In the given ray diagram, find the distance (in cm) between the two convex lenses.



Options:

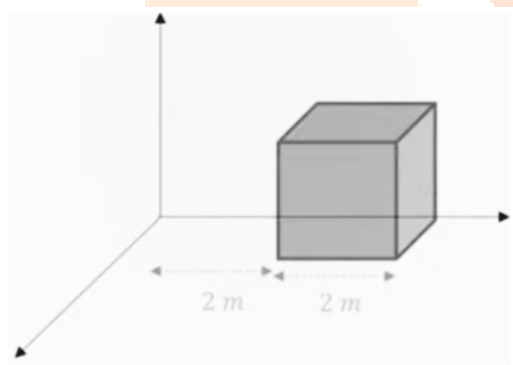
Answer: (25)

Question: Find the work done (in J) by force $F = 3x^2 + 2x - 5$ in moving a particle $x = 2$ to $x = 4$.

Options:

Answer: (58)

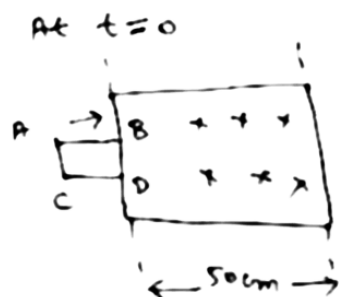
Question: There is an imaginary cube of side 2 m shown where edges are along axes. The electrostatic field varies as $\vec{E}(x) = 2x\hat{i}$, then flux through cube is Nm^2/C is ___



Options:

Answer: (16)

Question: Find the induced emf in the square loop of side 15 cm, moving with 2 cm/s after 10 sec.



Options:

(a) 0

- (b) 0.3 mV
- (c) 3 V
- (d) 9 V

Answer: (a)

Question: A spring exerts a force on a book $\vec{F} = -50\vec{x}$, where x is the change in length of the spring. Find the time period of oscillation. ($m = 0.5$ kg)



Options:

- (a) 0.63 s
- (b) 3.14 s
- (c) 1.57 s
- (d) 0.31 s

Answer: (a)

Question: A proton and a deuteron having same kinetic energy, enter a transverse uniform magnetic field. The radius of circular paths of proton and deuteron are in the ratio of

Options:

- (a) $\sqrt{2}$
- (b) $1/2\sqrt{2}$
- (c) $1/\sqrt{2}$
- (d) $2\sqrt{2}$

Answer: (c)

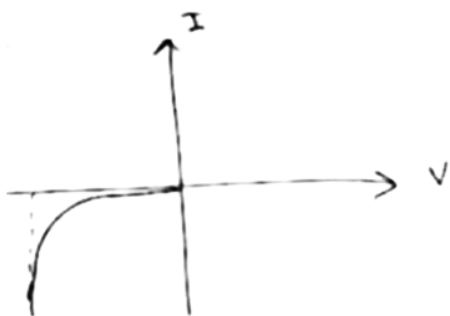
Question: A satellite of mass m is orbiting in an orbit of radius $2r$ from center of the planet of mass M and radius r . If satellite is given energy $E = \frac{GMm}{6r}$, then find new radius of orbit in which satellite will revolve

Options:

- (a) 14 r
- (b) 6 r
- (c) 8 r
- (d) 12 r

Answer: (b)

Question: For which of the following the I - V characteristic shown below is possible?

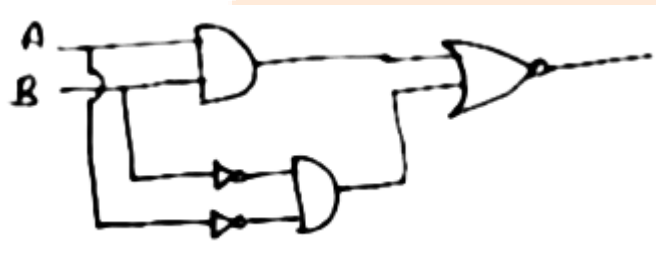


Options:

- (a) Transistor
- (b) Zener diode
- (c) Solar cell
- (d) Diode used as rectifies

Answer: (b)

Question: For the circuit shown, the truth table is given. Find x and y.



A	B	Y
0	0	0
0	1	x
1	0	y
1	1	0

Options:

- (a) 0, 0
- (b) 0, 1
- (c) 1, 0
- (d) 1, 1

Answer: (d)

Question: A ball of radius 10^{-4} m and density 10^5 kg/m³ is dropped from a height h into water (viscosity = 9.8×10^{-6} Pa - s) such that after falling into liquid, its speed does not change. Find the approximate value of height h .

Options:

- (a) 2200 m
- (b) 2350 m
- (c) 2470 m

(d) 2520 m

Answer: (c)

Question: If work function of a metal is 2.13 eV and energy per photon of incident light is 3.13 eV, then maximum kinetic energy of photoelectrons (in eV) will be ____.

Options:

Answer: (1)

Question: A photon of energy 10.2 eV is incident on hydrogen atom in ground state.

Thereafter, the number of lines emitted will be:

Options:

Answer: (2)



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Chemistry

Question: Correct order of bond angle of following compounds is :

Options:

- (a) $\text{BF}_3 > \text{PF}_3 > \text{ClF}_3$
- (b) $\text{PF}_3 > \text{ClF}_3 > \text{BF}_3$
- (c) $\text{ClF}_3 > \text{PF}_3 > \text{BF}_3$
- (d) $\text{BF}_3 > \text{ClF}_3 > \text{PF}_3$

Answer: (a)

Question: Identify the correct electronic configuration of Einsteinium is

Options:

- (a) $[\text{Rn}]5f^{14} 6d^1 7s^2$
- (b) $[\text{Rn}]5f^{11} 7s^2$
- (c) $[\text{Rn}]5f^{01} 6d^1 7s^2$
- (d) $[\text{Rn}]5f^{11} 6d^1 7s^1$

Answer: (b)

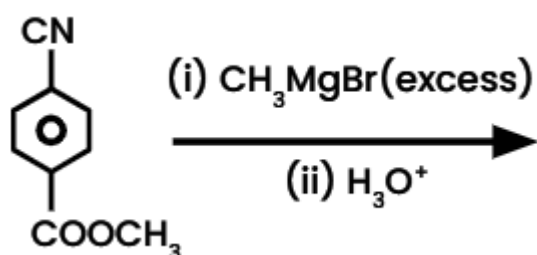
Question: Ca^{2+} makes which type of complex with EDTA

Options:

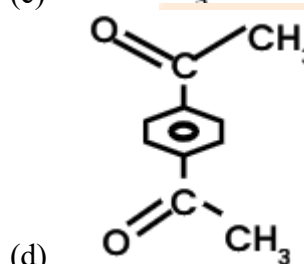
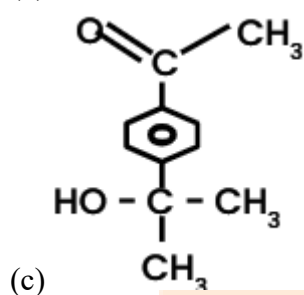
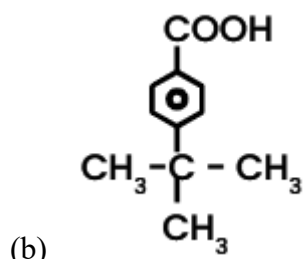
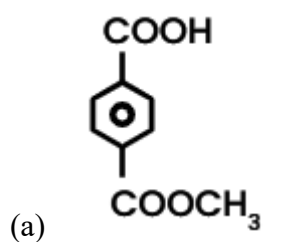
- (a) Trigonal bipyramidal
- (b) Square Planer
- (c) Tetrahedral
- (d) Octahedral

Answer: (d)

Question: The product obtained in the following reaction is :



Options:



Answer: (c)

Question: Fuming sulphuric acid has how many oxygen atoms ?

Options:

Answer: (7)

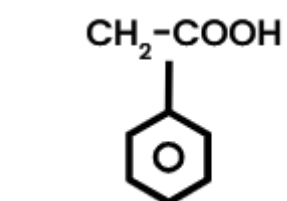
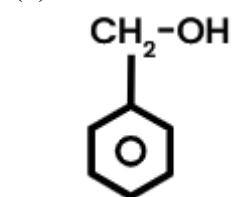
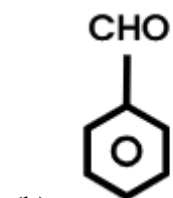
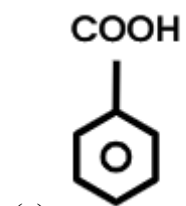
Question: Total sum of electrons in π^* orbitals of O_2 , O_2^+ , and O_2^- is :

Options:

Answer: (6)

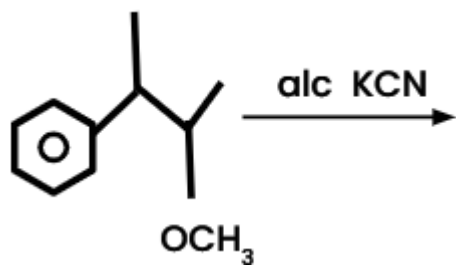
Question: Consider the following reaction and identify the major product P.

Options:

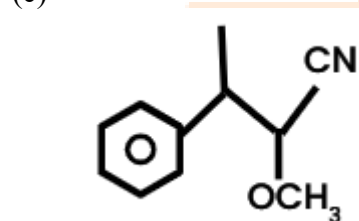
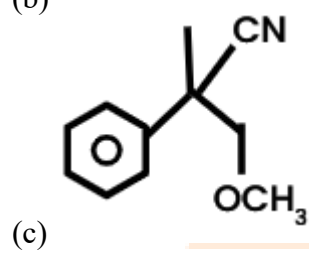
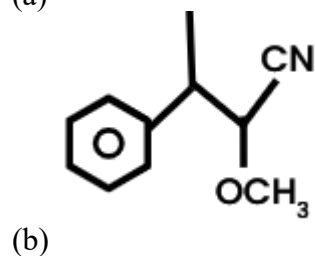
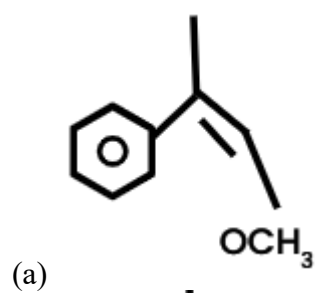


(d)
Answer: (a)

Question: Major product is :



Options:



(d)
Answer: (b)

Question: Which of the following is correct for strong electrolyte ($B > 0$) ?

Options:

(a) $\lambda_m - \lambda_m^0 - B\sqrt{C} = 0$

(b) $\lambda_m + \lambda_m^0 - B\sqrt{C} = 0$

(c) $\lambda_m - \lambda_m^0 + B\sqrt{C} = 0$

(d) $\lambda_m + \lambda_m^0 + B\sqrt{C} = 0$

Answer: (c)

Question: Which of the following will give positive iodoform test ?

Options:

(a) $\text{CH}_3\text{-CH}_2\text{-CH}_2\text{-CHO}$

(b) $\text{CH}_3\text{-CH-CH}_3$

|
OH

(c)

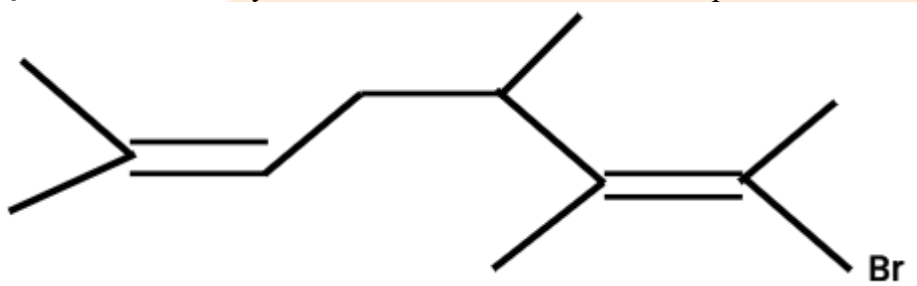
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$\text{CH}_3\text{-CH}_2\text{-C-CH}_2\text{-CH}_3$

(d) $\text{CH}_3\text{-CH}_2\text{-CH}_2\text{-CH}_2\text{-OH}$

Answer: (b)

Question: How many total number of stereoisomers are possible for the following structure ?



Options:

Answer: (4)

Question:

Match the list and choose the correct option.

List I

List II

(I) Ni-Cd cell

(a) Rechargeable

(II) Fuel cell

(b) Anode

(III) Mercury cell

(c) Used in hearing aid

(IV) Leclanche cell

(d) Combustion energy into electrical energy

Options:

(a) (i) - a ; (ii) -d; (iii)-c; (iv)-b

(b) (i) - b ; (ii) -a; (iii)-c; (iv)-d

(c) (i) - d ; (ii) -a; (iii)-c; (iv)-d

(d) (i) - a ; (ii) -b; (iii)-c; (iv)-d

Answer: (a)

Question: What is the correct order of C-C bond length of Ethane, Ethene and Ethyne?

Options:

- (a) Ethane > Ethene > Ethyne
- (b) Ethene > Ethane > Ethyne
- (c) Ethyne > Ethene > Ethane
- (d) Ethyne > Ethane > Ethane

Answer: (a)

Question:

Match the complexes given in List-I with hybridisation of central metal atom/ion given in List - II and choose the correct option

List I

List II

- | | |
|----------------------|---------------|
| (A) $K_2[Ni(CN)_4]$ | (a) sp^3 |
| (B) $[Ni(CO)_4]$ | (b) sp^3d^2 |
| (C) $[Co(NH_3)_6]Cl$ | (c) dsp^2 |
| (D) $Na_3[CoF_6]$ | (d) d^2sp^3 |

Options:

- (a) A - I, B - II, C - III, D - IV
- (b) A - III, B - I, C - IV, D - II
- (c) A - IV, B - III, C - II, D - I
- (d) A - I, B - II, C - IV, D - III

Answer: (b)

Question: Which one of the following statements regarding glucose is incorrect

Options:

- (a) Glucose is one of the monosaccharides of sucrose
- (b) Glucose dissolves in water because it has aldehyde group
- (c) Glucose has six carbon atoms in its structure
- (d) Glucose is an aldose

Answer: (b)

Question: Which of the following compounds do not give Tollen's Test?

Options:

- (a) Formaldehyde
- (b) Formic acid
- (c) Benzaldehyde
- (d) Acetone

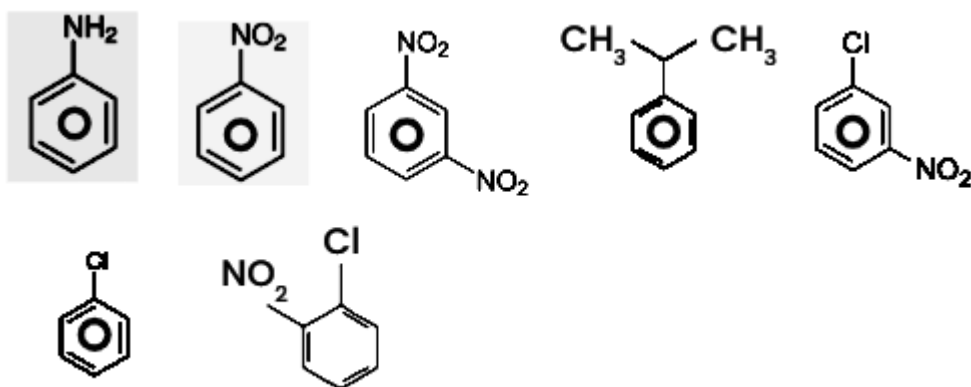
Answer: (d)

Question: Among the elements - Sc, Ti, V, Cr and Mn; find magnetic moment of element which have highest ionization enthalpy in +2 oxidation state (Nearest integer)

Options:

Answer: (6)

Question: How many of the following compounds will give Friedel Craft's reaction ?



Options:

Answer: (3)

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[EVENINGSHIFT]

Maths

Question: If $\frac{z-2i}{z+2i}$ is purely imaginary then find the maximum value of $|z+8+6i|$.

Options:

- (a) 11
- (b) 12
- (c) 14
- (d) 16

Answer: (d)

Question: Variance of the following is 160

f	2	1	1	1	1	1
x	C	2C	3C	4C	5C	6C

Find c

Answer: (c=7)

Question: $\sum_{n=0}^{\infty} ar^n = 57$ $\sum_{n=0}^{\infty} a^3 r^{3n} = 9747$ What is $a + 18r = ?$

Answer: (31)

Question: $\lim_{x \rightarrow 0} \frac{e - (1+2x)^{1/2x}}{x}$

Options:

- (a) e
- (b) 1/e
- (c) e^2
- (d) $1/e^2$

Answer: (a)

Question: In the expansion of $\left(x^{2/3} + \frac{1}{2}x^{-2/5}\right)^9$ find the sum of coefficients of $x^{2/3}$ and $x^{-2/5}$

Options:

- (a) 6.75
- (b) 5.25
- (c) 3.28
- (d) 9.76

Answer: (b)

Question: $\lim_{x \rightarrow \frac{\pi}{2}} \frac{\int x^{\left(\frac{\pi}{2}\right)^3} (\sin(2x) + \cos x) dx}{\left(x - \frac{\pi}{2}\right)}$

Answer: (2)

Question: $2\sin^{-1}(x) + 3\cos^{-1}(x) = \frac{7\pi}{5}$, find number of real solution of equation

Answer: (No real solution)

Question: $I = \int_{-1}^2 \ln(x + \sqrt{1+x^2}) dx$

Answer: (1)

Question: $I = \int_{-1}^2 \ln(x + \sqrt{1+x^2}) dx$

Answer: (1)

Question: $\int_{\frac{1}{4}}^{\frac{3}{4}} \cos\left(\frac{2\cot^{-1}\sqrt{\frac{1-x}{1+x}}}{\theta}\right) dx$

Answer: (0)

$\frac{x^2}{100} + \frac{y^2}{75} = 1$

Question: $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

Answer: (225)

Question: $f'(x) = 3f(x) + x$ and $\lim_{x \rightarrow \infty} f(x) = 7$; $\lim_{x \rightarrow 0} f(x) = 1$. Find $9f(-\ln 3) = ?$

Answer: (61)

Question: If $f'(x) = 3f(x) + x$ and $f(0) = 1$, then $f(x)$ is:

Options:

(a) $-\frac{x}{3} + \frac{10}{9}e^{-3x}$

(b) $-\frac{x}{3} - \frac{1}{9} + \frac{10}{9}e^{3x}$

(c) $-\frac{x}{3} - \frac{10}{9}e^{-3x}$

(d) $-\frac{x}{3} - \frac{1}{9} + \frac{10}{9}e^{2x}$

Answer: (b)

Question: Dice is thrown 3 times then find the probability that $x_1 < x_2 < x_3$ (there $x_1, x_2, x_3 \in [1, 6]$) where x_1, x_2, x_3 are outcomes on dice.

Options:

(a) $7/54$

(b) $5/54$

(c) $11/54$

(d) $17/54$

Answer: (b)

Question: Find the area bounded by ellipse $x^2 + 3y^2 = 18$ below the line $y = x$ (in first quadrant)

Options:

(a) $3\pi + 1$

(b) $\sqrt{3}\pi$

(c) $3\pi - \frac{3}{4}$

(d) $3\pi + \frac{1}{4}$

Answer: ()

